

VI.2 Stationary Phase Method: Calculation of the First Terms

The stationary phase method proceeds in the same way. We want to evaluate an integral of the form

$$I = \int M(x) e^{i f(x)} dx \quad (\text{B.9})$$

under the assumption that the relative variations of $M(x)$ are slow compared to those of $f(x)$. Thus, I is a rapidly oscillatory integral, so one can expect compensation phenomena on the intervals in x where $e^{i f'(x)}$ completes one full oscillation. This is made concrete by the following result:

Lemma 1. *Let $M \in C_c^\infty(\mathbb{R})$ and let $f \in C^\infty(\mathbb{R})$ be a real-valued function such that $f'(x) > 0$ for all x in the support of M . Then*

$$\int M(x) e^{i \mu f(x)} dx = O(\mu^{-k}) \quad \forall k \in \mathbb{N}. \quad (\text{B.10})$$

Proof: Suppose $x \mapsto f$ is a change of variable on $\text{Supp}(M)$. If this is not the case, we reduce to it by using a partition of unity. More precisely, we write

$$M(x) = \sum M(x) u_\lambda(x),$$

where the $u_\lambda \in C_c^\infty(\mathbb{R})$ form a partition of unity, i.e. $\sum_\lambda u_\lambda(x) = 1$ almost everywhere, and for each λ , $x \mapsto f$ is a change of variable on $\text{Supp}(M u_\lambda)$. The integral then becomes

$$\int \frac{M[x(f)]}{f'} e^{i \mu f} df,$$

i.e. it is the Fourier transform of M/f' . The lemma is then obtained by k integrations by parts. Here, the parameter μ was introduced in order to control the oscillations of the argument of the exponential. The limit $\mu \rightarrow \infty$ is called the asymptotic limit.

Next, consider the case where there is one or more points x_s (called stationary points) such that $f'(x_s) = 0$. The compensation mechanism no longer operates in the neighborhood of these points, which contribute in an important way to I . It is therefore natural to construct a series expansion around each stationary point. Let us write

$$I = e^{i f(x_s)} \int M(x) e^{i [f(x) - f(x_s)]} dx. \quad (\text{B.11})$$

Suppose for simplicity that $f''(x_s) > 0$. Then $(f(x) - f(x_s))$ behaves like $(x - x_s)^2$ near x_s . Introduce a variable u such that

$$-u^2 = i [f(x) - f(x_s)]. \quad (\text{B.12})$$

Near x_s , one obtains

$$u^2 \approx -\frac{i}{2} (x - x_s)^2 f''(x_s). \quad (\text{B.13})$$

We then write

$$I = e^{i f(x_s)} \int \frac{M(x(u))}{\frac{dx}{du}} e^{-u^2} du, \quad (\text{B.14})$$

which we expand into a series:

$$I = e^{i f(x_s)} \sum_k \frac{M^{(k)}(0)}{k!} \int u^k e^{-u^2} du \quad (\text{B.15})$$

$$= e^{i f(x_s)} \sum_k \frac{M^{(2k)}(0)}{(2k)!} (k - \frac{1}{2})! , \quad (\text{B.16})$$

where we have set

$$M(u) = \frac{M(x(u))}{\frac{dx}{du}}.$$

Consider, for instance, the first term of this series expansion, which is the one we shall use essentially in what follows:

$$M(0) = \frac{M(x_s) \sqrt{2}}{|f''(x_s)|^{1/2}} e^{\frac{i}{4} \pi \text{sgn}[f''(x_s)]},$$

yielding

$$I_{(0)} = \sqrt{\frac{\pi}{2}} \frac{e^{\frac{i}{4} \pi \text{sgn}[f''(x_s)]}}{|f''(x_s)|^{1/2}} M(x_s) e^{i f(x_s)}. \quad (\text{B.17})$$

The next term is proportional to $\frac{dM}{du}(0)$, which is finite, and multiplied by $\int_{-\infty}^{\infty} u e^{-u^2} du = 0$; hence $I_{(1)} = 0$. Moving to the second-order term, a straightforward calculation shows that

$$\frac{d^2 M(u)}{du^2} = M'' u^2 - 3 M' u' u'' + M(3 u' u^2 - u' u''') / u^5, \quad (\text{B.18})$$

and evaluating the successive derivatives of u at x_s leads to

$$\begin{aligned} \left. \frac{d^2 M(u)}{du^2} \right|_0 &= 2\sqrt{2} \frac{e^{\frac{3i}{4}\pi \operatorname{sgn}[f''(x_s)]}}{|f''(x_s)|^{3/2}} \left\{ M''(x_s) f''(x_s) f'''(x_s)^2 \right\}, \\ &- M'(x_s) f''(x_s) f'''(x_s) + \frac{M(x_s)}{12} [5 f'''(x_s)^2 - 3 f'''(x_s) f''(x_s)] \Big\}, \quad (\text{B.19}) \end{aligned}$$

so that the third term is given by

$$\begin{aligned} I_{(2)} &= \frac{\sqrt{\pi}}{4} \frac{e^{\frac{3i}{4}\pi \operatorname{sgn}[f''(x_s)]}}{[f''(x_s)]^{3/2} |f''(x_s)|^{3/2}} e^{if(x_s)} \left\{ M'(x_s) f''(x_s) f'''(x_s) \right. \\ &\quad \left. - \frac{M(x_s)}{12} [(5 f'''(x_s))^2 - 3 f'''(x_s) f''(x_s)] \right\}. \quad (\text{B.20}) \end{aligned}$$

Remark: A fully similar procedure can be applied if, for example, all the derivatives of $f(x_s)$ vanish up to some order $k-1$ and $f^{(k)}(x_s) > 0$. In these conditions, the asymptotic series computation reduces to evaluating integrals of the type

$$\int u^n e^{-u^k} du.$$

As an example, the leading term of the asymptotic series in the case $k=3$ is written:

$$6^{1/3} \left(\frac{1}{3}\right)^{1/3} \frac{e^{\frac{i}{4}\pi \operatorname{sgn}[f^{(3)}(x_s)]}}{|f^{(3)}(x_s)|^{1/3}} M(x_s) e^{if(x_s)}.$$

VI.3 Stationary Phase Method: Quality of the Approximation

In applications of the stationary phase method that we develop, we shall limit ourselves to the approximation given by the first term (B.17) of the asymptotic expansion. To judge the quality of the approximation, one may either compare it to the result obtained with the next term (B.19) in the expansion, or attempt to estimate the error committed. We give here a result that allows one to estimate this error. The interested reader may refer to [?] for a more detailed presentation and more precise estimates.

Theorem 1. *Let $M \in C_c^2(\mathbb{R})$ and $f \in C^4(\mathbb{R})$ be real-valued. Suppose x_s is the unique stationary point of $f(x)$ and that $f''(x)$ does not vanish on the support of M . Then:*

1. *There exists a constant $C > 0$ such that*

$$\int M(x) e^{i f(x)} dx = \sqrt{\frac{2\pi}{|f''(x_s)|}} \exp\left\{\frac{i\pi}{4} \operatorname{sgn}[f''(x_s)]\right\} M(x_s) e^{i f(x_s)} + r, \quad (\text{B.21})$$

$$|r| \leq C \sup_{x \in \operatorname{Supp} M} \left[\frac{|M'(x) f''(x)|}{|f''(x)|^{3/2}} + \frac{|M'(x) f'''(x)|}{|f''(x)|^{5/2}} + \frac{|M(x) f''''(x)|}{|f''(x)|^{5/2}} \right]. \quad (\text{B.22})$$

2. *In the limit $\mu \rightarrow \infty$, we have the asymptotic behavior*

$$\int M(x) e^{i \mu f(x)} dx - \sqrt{\frac{2\pi}{\mu |f''(x_s)|}} \exp\left\{\frac{i\pi}{4} \operatorname{sgn}[f''(x_s)]\right\} M(x_s) e^{i \mu f(x_s)} = O(\mu^{-3/2}). \quad (\text{B.23})$$

Proof. 1) If $x_0 \in \mathbb{R}$, if U is a neighborhood of x_0 and if $f \in C^k(U)$, then Taylor's formula states:

$$f(x_0 + \epsilon) = \sum_{j=0}^{k-1} \frac{\epsilon^j}{j!} f^{(j)}(x_0) + \frac{\epsilon^k}{(k-1)!} \int_0^1 (1-t)^{k-1} f^{(k)}(x_0 + t\epsilon) dt. \quad (\text{B.24})$$

For any $x = x_0 + \epsilon \in M$, with $k = 2$, we have $f(x_0 + \epsilon) = f(x_0) + \epsilon f'(x_0) + \epsilon^2 F_2(x)$, where $F_2 \in C^1(U)$ and $|F_2(x)| \leq \frac{1}{2} \sup_{x \in U} |f''(x)|$. Inserting this into (B.14) and using the evaluations of the first terms $I_{(0)}$ and $I_{(1)}$ proves (B.21). We get:

$$r = \frac{1}{2} e^{i f(x_s)} \int F_2(u) u^2 e^{-u^2} du, \quad (\text{B.25})$$

and

$$|r| \leq \frac{1}{2} \sup_{z \in \mathbb{R}} |F_2(x)| \int u^2 e^{-u^2} du.$$

Combining this estimate with a similar calculation for (B.18) and (B.19) leads to (B.22).

2) The statement about $\mu \rightarrow \infty$ in (2) follows trivially from (1), and the theorem is proved.

Notation: Throughout this chapter, we use the symbol $A \approx B$ to signify that B is the first term of the asymptotic expansion of A .